TEXORA · LECTURE SHEET

The Combing Machine

Combing removes short fibres as noil and delivers the most parallel, even sliver in spinning — at the cost of the largest single waste stream in the mill. This sheet works the preparation calculation (lap building), noil percentage, production per head and the comber's overall draft.

YM-3207 · ADVANCED SPINNING CALCULATIONS · TOPIC 3 · SPECIALIZATION — YARN MANUFACTURING

1. Why comb, and why it needs a prepared lap

A comber works on the ends of a fibre fringe. It grips a lap fringe with nippers, combs the free end with a rotating half-lap of needles, draws the combed fringe through a top comb into the previously detached web, and ejects the short fibres, neps and trash left behind as **noil**. The result is a sliver with a higher mean fibre length, far fewer short fibres, and fibres lying straighter and more parallel than carding alone can give — the basis of fine, strong, smooth combed yarns.

A single card sliver is too uneven in cross-section to feed directly: thick and thin places would be gripped unevenly. Preparation therefore doubles many slivers into a **lap** of even weight, usually on a sliver-lap or ribbon-lap machine, sometimes with a drawframe passage before it.

WORKED EXAMPLE — BUILDING THE COMBER LAP

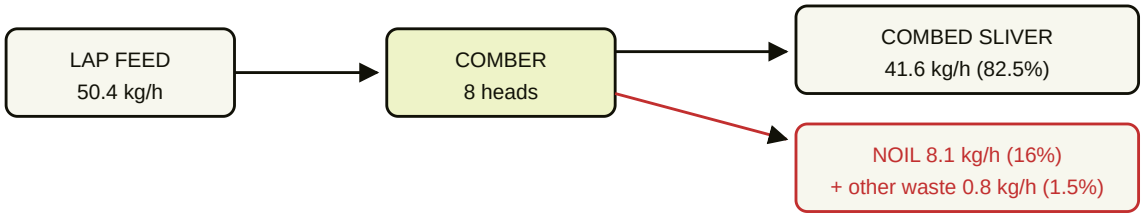
A ribbon-lap machine doubles 16 card slivers of 3.94 g/m each with a draft of 1.05.
Lap weight = 16 × 3.94 / 1.05 = **60.0 g/m**.
The draft close to 1 keeps the combined sheet from being stretched thin again; the purpose of the machine is doubling (evenness), not drafting. This 60 g/m lap is the feed used in all the comber calculations below.

2. Noil percentage — the comber's defining number

$$\text{Noil (\%)} = \text{mass of noil extracted} / \text{mass of lap fed} \times 100$$

Noil is set by the machine's geometry and timing — principally the feed per nip and the detaching (gauge) setting. Shorter feed per nip combs each fibre fringe through more cycles per unit length, so more short fibre is extracted and noil percentage rises; a wider detaching setting also lets the nipper

hold the fringe further back, releasing more short fibre into the noil. Fine counts need more short fibre removed (noil 18–25%); coarse combed counts may run at 10–14%.



Mass balance of the worked example: 50.4 = 41.6 + 8.1 + 0.8 kg/h (to rounding).

Figure 1 — Comber mass flow for the worked example: 16% noil and 1.5% other waste leave 82.5% of the feed as combed sliver.

3. Production of the comber

Each head feeds a fixed length of lap per nip (combing cycle), so feed rate follows directly from nips per minute.

$$\text{Feed per head (m/min)} = \text{nips per minute} \times \text{feed per nip (mm)} / 1000$$

WORKED EXAMPLE — COMBER PRODUCTION

Nips = 350 per minute; feed per nip = 5.0 mm; lap = 60.0 g/m; machine has 8 heads; noil = 16%, other waste = 1.5%.

Feed per head = 350 × 5.0 / 1000 = 1.75 m/min.

Feed mass per head = 1.75 × 60.0 g = 105 g/min = 6.30 kg/h.

Feed, whole machine = 6.30 × 8 = **50.4 kg/h**.

Delivered sliver = 50.4 × (1 − 0.16 − 0.015) = 50.4 × 0.825 = **41.6 kg/h**, i.e. 5.20 kg/h per head.

Delivered sliver weight is 3.9 g/m, so delivery speed per head = 5200 g/h ÷ 60 ÷ 3.9 g/m = **22.2 m/min**.

4. Overall draft at the comber

Because roughly a sixth of the feed leaves as noil, the length delivered is much greater than the length fed — the comber is also a drafting machine:

$$\text{Overall draft} = \text{delivered length} / \text{fed length} = 22.2 / 1.75 = \mathbf{12.7}$$

This draft is not set by change wheels as on a drawframe; it is the outcome of feed per nip, noil level and the sliver weight the spinner chooses to deliver. If the delivered sliver is too heavy or light, the correction is made by changing lap weight or feed per nip — and every such change also moves the noil percentage, which is why comber calculations are always done as a set (see YM-4101, Topic 6, for the control side).

5. What combing buys — and costs

GAINS	COSTS
Short fibres removed: mean length rises, length distribution narrows	Noil of 10–25% of feed — the largest waste stream in spinning
Neps and fine trash largely removed with the noil	Extra preparation machines (lap formers, often a drawframe)
Fibres straighter and more parallel: stronger, smoother, more lustrous yarn	Low production per machine (41.6 kg/h here) against high capital cost
Finer counts spinnable from a given cotton	Noil must be sold or reprocessed at lower value

Glossary

Noil — the short fibres, neps and trash extracted by the comber. **Nip** — one combing cycle of a head. **Feed per nip** — the length of lap the feed roller advances each cycle. **Detaching setting** — the distance between the nipper grip and the detaching rollers at their closest approach; a primary noil control. **Half-lap** — the needle-covered segment of the circular comb. **Top comb** — the stationary comb that cleans the trailing end of the fringe during detachment. **Ribbon lap** — the lap built by doubling slivers for comber feed.

Key points

• Preparation doubles 16 slivers into a 60.0 g/m lap in the example (draft 1.05).
• Noil % is set mainly by feed per nip and detaching setting: shorter feed → more noil.
• Production per head = nips × feed per nip × lap weight; the example machine feeds 50.4 kg/h and delivers 41.6 kg/h.
• Overall comber draft = delivered length/fed length = 12.7 in the example.
• Combing trades a large, valuable waste stream for length uniformity, cleanliness and fineness.

Practice

1. A comber runs at 300 nips/min with a feed of 5.4 mm per nip, lap 62 g/m, 8 heads, noil 14% and other waste 1%. Calculate the feed rate and the delivered sliver production of the machine.
2. A ribbon-lap machine doubles 20 slivers of 3.6 g/m with draft 1.08. Calculate the lap weight.
3. A mill wants to raise noil from 14% to 17% without changing lap weight. State two machine adjustments that move noil in that direction and one consequence for production.

Answer key

1. Feed per head = 300 × 5.4/1000 = 1.62 m/min. Feed mass = 1.62 × 62 = 100.4 g/min = 6.03 kg/h per head; machine feed = 6.03 × 8 = **48.2 kg/h**. Delivered = 48.2 × (1 − 0.14 − 0.01) = 48.2 × 0.85 = **41.0 kg/h**.
2. Lap = 20 × 3.6 / 1.08 = **66.7 g/m**.
3. Reduce the feed per nip (each fringe is combed through more cycles per unit length), and/or close the detaching setting so the fringe is held further back. Consequence: delivered sliver production falls roughly in proportion to the extra noil removed (about 3% more of the feed leaves as waste), and raw material cost per kilogram of yarn rises.

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